

Single Inheritance

Basic Information

Title: Single Inheritance

File Name: SingleInheritanceDemo.java

Objective / Aim

To demonstrate single inheritance in Java, where one child class inherits from one parent class.

System Requirements

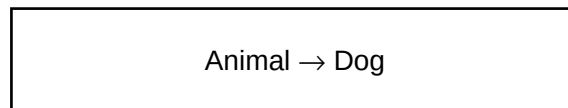
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the parent class Animal with eat().
3. Create Dog extending Animal.
4. Create an object of Dog.
5. Call the inherited eat() method and Dog's bark() method.
6. End the program.

UML Class Diagram



Source Code

```
// Parent class
class Animal {

    void eat() {
        System.out.println("Animal is eating");
    }
}

// Child class inherits Animal
class Dog extends Animal {

    void bark() {
        System.out.println("Dog is barking");
    }
}

public class SingleInheritanceDemo {
    public static void main(String[] args) {

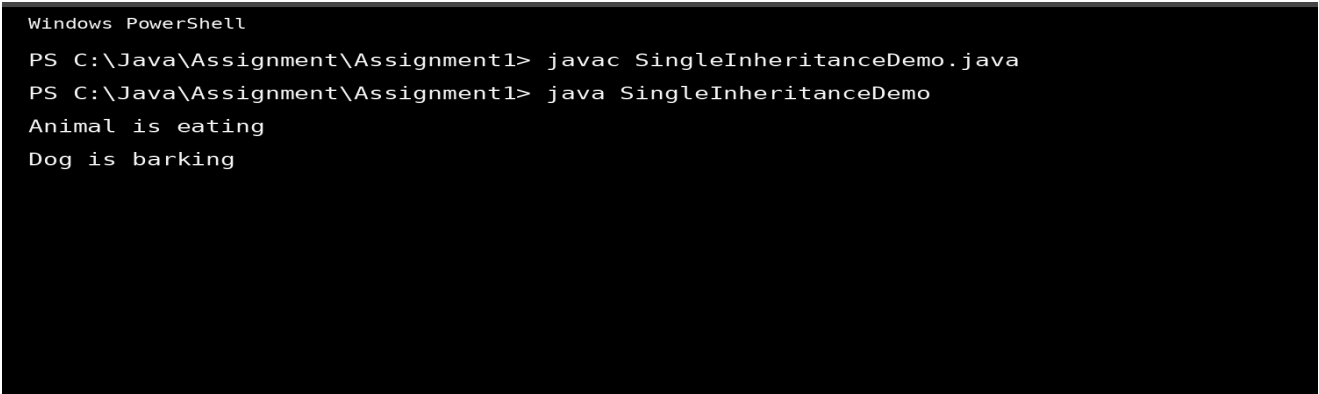
        // Create object of child class
        Dog dog = new Dog();

        dog.eat();
        dog.bark();
    }
}
```

Compilation & Execution Commands

```
javac SingleInheritanceDemo.java
java SingleInheritanceDemo
```

Output Screen



```
Windows PowerShell
PS C:\Java\Assignment\Assignment1> javac SingleInheritanceDemo.java
PS C:\Java\Assignment\Assignment1> java SingleInheritanceDemo
Animal is eating
Dog is barking
```

Result

Thus, the Single Inheritance program was implemented successfully using Java inheritance concepts.